
Sentiment-Driven Cryptocurrency Trading: Integrating Naïve Bayes and Agentic AI Orchestration for Enhanced Signal Generation

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Abstract

Cryptocurrency markets, characterised by extreme volatility and the absence of conventional fundamental anchors, present a compelling domain for sentiment-driven algorithmic trading. Existing approaches relying on static probabilistic classifiers — such as the Naïve Bayes Bernoulli model formalised by Kakushadze and Serur (2018) — offer interpretability and computational efficiency but lack the adaptive capacity required in dynamic, multi-source information environments. This paper proposes the Naïve Bayes Agentic AI (NB-AAI) framework: a novel hybrid architecture integrating the Naïve Bayes Bernoulli model with a six-agent orchestration system for enhanced cryptocurrency trading signal generation.

Applied to Bitcoin (BTC/USD) and Ethereum (ETH/USD) over the 2020–2024 period, the framework deploys a data collection agent, a Naïve Bayes classification agent, a FinBERT LLM validation agent, a signal generation agent, a risk management agent, and an orchestrating agent. On a 36-month out-of-sample test spanning four market regimes — the 2022 Crypto Winter, two recovery phases, and the 2024 bull run — NB-AAI achieves a BTC Sharpe ratio of 1.47, an annualised return of +52.6% net of costs, and a maximum drawdown of -22.4%, versus 0.31, +18.4%, and -77.1% for passive buy-and-hold. All improvements over five comparison baselines are statistically confirmed by the Diebold-Mariano test ($p < 0.001$). An ablation study reveals superadditivity of the full pipeline over individual components. The framework is released as open-source code to support replication.

Keywords: Naïve Bayes; Sentiment Analysis; Cryptocurrency Trading; Agentic AI; Multi-Agent Systems; FinBERT; Bitcoin; Signal Generation

1. INTRODUCTION

1.1 Background and Motivation

The global cryptocurrency market has grown from a niche technological experiment into a multi-trillion-dollar asset class. Unlike traditional equity markets, cryptocurrencies such as Bitcoin (BTC) and Ethereum (ETH) possess no intrinsic fundamental anchors — no earnings, dividends, or book value. As Nakamoto (2008) first conceived, these assets derive legitimacy purely from decentralised consensus. Consequently, price dynamics are disproportionately governed by collective market sentiment and information asymmetries propagated through social media and digital news ecosystems.

Bollen et al. (2011) provided seminal evidence that public mood expressed on Twitter predicts equity market movements, opening a productive line of inquiry into text-based financial forecasting. Sentiment signals derived from social media are especially potent in cryptocurrency markets, where the absence of fundamental constraints renders prices highly susceptible to narrative-driven momentum (Kraaijeveld & De Smedt, 2020). Studies report directional accuracy improvements of 10-15 percentage points over baseline models when sentiment features are incorporated (Abraham et al., 2018; Garcia & Schweitzer, 2015).

Kakushadze and Serur (2018) formalised a Naïve Bayes Bernoulli framework for social media-based BTC direction forecasting. Given N binary feature vectors X_i from tweet T_i , the posterior class probability follows Bayes' theorem:

$$P(A | B) = P(B | A) \cdot P(A) / P(B) \quad (1)$$

The joint posterior over class C_α given all features, under the Naïve conditional independence assumption, simplifies to (Kakushadze & Serur, 2018, Eq. 541):

$$P(C_\alpha | X_1, \dots, X_N) = \gamma \cdot P(C_\alpha) \cdot \prod_i P(X_i | C_\alpha) \quad (2)$$

For the Bernoulli model, each marginal factor is:

$$P(X_i | C_\alpha) = \prod_{a=1}^{M_a} Q_{ia} \alpha \quad (3)$$

where $Q_{ia} \alpha = P(w_a | C_\alpha)$ if $X_{ia}=1$, and $1-P(w_a | C_\alpha)$ if $X_{ia}=0$. The classification rule is (Kakushadze & Serur, 2018, Eq. 546):

$$C_{pred} = \operatorname{argmax}_\alpha P(C_\alpha) \cdot \prod_i \prod_a [P(w_a | C_\alpha)]^{X_{ia}} \cdot [1 - P(w_a | C_\alpha)]^{(1 - X_{ia})} \quad (4)$$

1.2 Research Gap

Three structural limitations constrain the Kakushadze-Serur framework in modern environments:

Gap 1	Static Vocabulary: Vocabulary V constructed offline becomes stale as cryptocurrency discourse evolves, degrading out-of-sample accuracy (Kraaijeveld & De Smedt, 2020).
Gap 2	Naïve Independence: Cannot capture sarcasm, negation, and contextual nuance — phenomena prevalent in social media (Araci, 2019; Devlin et al., 2019).
Gap 3	Single-Source Architecture: Single data stream implementations miss the signal quality gains from multi-source fusion (Fang et al., 2022).

These gaps, compounded by the absence of real-time adaptive coordination, define the primary contribution of this paper.

1.3 Research Contributions

This paper makes four original contributions:

- 1 The NB-AAI hybrid framework: the first integration of Naïve Bayes Bernoulli classification within a six-agent Agentic AI pipeline for cryptocurrency trading.
- 2 Multi-regime empirical evaluation on BTC/USD and ETH/USD (2020-2024) spanning four market regimes.
- 3 A systematic ablation study quantifying the marginal contribution of each of the six agents.
- 4 A reproducible open-source Python implementation with Docker container and Colab notebooks.

1.4 Research Hypotheses

H1	Naïve Bayes sentiment classification generates statistically superior returns to buy-and-hold on BTC/USD and ETH/USD (2022-2024).
H2	FinBERT LLM validation of Naïve Bayes outputs yields a statistically significant improvement in Sharpe ratio over standalone Naïve Bayes.
H3	The full NB-AAI pipeline achieves superior risk-adjusted performance over any single-component baseline, across all four market regimes.

1.5 Paper Structure

Section 2 reviews the literature. Section 3 develops the theoretical framework. Section 4 describes the methodology. Section 5 reports results. Section 6 presents future research directions. Section 7 concludes.

2. LITERATURE REVIEW

This section surveys five bodies of literature: (i) sentiment analysis in financial markets, (ii) the Naïve Bayes framework in trading, (iii) cryptocurrency market structure, (iv) large language models in finance, and (v) agentic AI systems.

2.1 Sentiment Analysis in Financial Markets

Tetlock (2007) demonstrated that pessimism in Wall Street Journal columns significantly predicts next-day Dow Jones returns. Bollen et al. (2011), analysing 9.7 million tweets, found public mood states Granger-cause DJIA values with 87.6% directional accuracy. The methodological evolution proceeded from dictionary-based lexicons (Loughran & McDonald, 2011) through machine learning classifiers (Pang et al., 2002) to contextualised language models (BERT; Devlin et al., 2019). For cryptocurrency markets, Kraaijeveld and De Smedt (2020) found that sentiment from highly followed accounts exhibits stronger predictive power, underpinning the source-weighting design of NB-AAI's Signal Fusion Agent.

2.2 The Naïve Bayes Framework in Trading

Despite the conditional independence assumption's technical violation in real-world data, Naïve Bayes classifiers achieve competitive accuracy in high-dimensional text classification tasks (Rish, 2001). Lamon et al. (2017) reported 77.4% directional accuracy on BTC using Naïve Bayes — exceeding SVM at 73.8% — attributing this to the high-noise, short-text nature of social media. The critical limitation, acknowledged by both Lamon et al. (2017) and Kakushadze and Serur (2018), is the static vocabulary, which NB-AAI's adaptive Orchestration Agent is specifically designed to overcome.

2.3 Cryptocurrency Market Structure and Trading Strategies

Urquhart (2016) documented significant return predictability in early Bitcoin markets. Fang et al. (2022), surveying 146 papers, identified sentiment analysis as one of three dominant signal paradigms, outperforming pure price-momentum strategies on risk-adjusted metrics. Otabek and Choi (2024) demonstrated that deep Q-network agents achieve superior Sharpe ratios to buy-and-hold on BTC, though lacking the interpretability preserved by NB-AAI's Bayesian core.

2.4 Large Language Models in Financial Applications

Araci (2019) introduced FinBERT — fine-tuned on 1.8 million financial news articles — achieving a 15.9% F1-score improvement over general BERT for financial sentiment classification. Lopez-Lira and Tang (2023) demonstrated GPT-4 sentiment scores predict next-day stock returns beyond human annotators. The key limitation of standalone LLM approaches — GPU inference latency — is addressed in NB-AAI by routing only low-confidence documents (approximately 21% of volume) to FinBERT.

2.5 Agentic AI Systems and Multi-Agent Orchestration

Park et al. (2023) demonstrated emergent social behaviours in LLM-based agent ensembles — information diffusion and norm formation — directly analogous to the market sentiment dynamics NB-AAI exploits. Wu et al. (2023) introduced AutoGen for multi-agent conversation frameworks; LangGraph (LangChain, 2024) provides the state-machine infrastructure adopted in this paper. Zhang et al. (2024) proposed FinAgent for stock prediction but without Naïve Bayes integration or cryptocurrency application — the precise gap NB-AAI fills.

2.6 Summary and Research Positioning

Table 1. Comparison of Selected Prior Works with the Proposed NB-AAI Framework

Study	Naïve Bayes	LLM	Multi-Agent	Crypto Market
Bollen et al. (2011)	✓	✗	✗	✗ (Equity)
Lamon et al. (2017)	✓	✗	✗	✓
Kraaijeveld & De Smedt (2020)	✗	✗	✗	✓
Araci / FinBERT (2019)	✗	✓	✗	✗
Khedr et al. (2021)	✗	✓	✗	✓
Li et al. (2023)	✗	✓ (partial)	✓	✗ (Equity)
Zhang et al. / FinAgent (2024)	✗	✓	✓ (partial)	✗ (Equity)
Kakushadze & Serur (2018)	✓	✗	✗	✓
NB-AAI (This Paper)	✓	✓	✓	✓

NB-AAI is the only framework that simultaneously integrates all four components, occupying a unique position in the literature landscape.

3. THEORETICAL FRAMEWORK

3.1 The Naïve Bayes Bernoulli Classifier

3.1.1 Problem Setup

Let $D = \{T_1, \dots, T_N\}$ be a corpus of N social media documents. Each document T_i is assigned class label $y_i \in \{+1, -1\}$, where $+1$ denotes bullish (C_+) and -1 denotes bearish (C_-). A vocabulary $V = \{w_1, \dots, w_M\}$ of M keywords is constructed via TF-IDF. Each document T_i is represented as:

$$X_i = (X_{i1}, X_{i2}, \dots, X_{iM}) \in \{0, 1\}^M \quad (5)$$

where $X_{ia}=1$ if keyword w_a appears in T_i , and 0 otherwise.

3.1.2 Bayesian Derivation

Applying Bayes' theorem to compute the class posterior:

$$P(C_\alpha | X_i) = P(X_i | C_\alpha) \cdot P(C_\alpha) / P(X_i) \quad (6)$$

Classification reduces to the argmax over classes (denominator is class-invariant):

$$C_{pred} = \operatorname{argmax}_\alpha P(C_\alpha) \cdot P(X_i | C_\alpha) \quad (7)$$

Under the Naïve independence assumption, the joint likelihood factorises:

$$P(X_i | C_\alpha) = \prod_{a=1}^M P(X_{ia} | C_\alpha) \quad (8)$$

For the Bernoulli model each marginal is:

$$P(X_{ia} | C_\alpha) = P(w_a | C_\alpha)^{X_{ia}} \cdot [1 - P(w_a | C_\alpha)]^{(1-X_{ia})} \quad (9)$$

Combining Eqs. (7)-(9) yields the full decision rule (Kakushadze & Serur, 2018, Eq. 546):

$$C_{pred} = \operatorname{argmax}_\alpha P(C_\alpha) \cdot \prod_i \prod_a [P(w_a | C_\alpha)]^{X_{ia}} \cdot [1 - P(w_a | C_\alpha)]^{(1-X_{ia})} \quad (10)$$

In practice evaluated in log-space to prevent numerical underflow:

$$\log P(C\alpha|X_i) \propto \log P(C\alpha) + \sum_a [X_{ia} \cdot \log P(w_a|C\alpha) + (1-X_{ia}) \cdot \log(1-P(w_a|C\alpha))] \quad (11)$$

3.2 Laplace Smoothing

To prevent zero-probability estimation for unseen vocabulary terms:

$$\hat{P}(w_a | C\alpha) = (\text{count}(w_a, C\alpha) + \alpha) / (N\alpha + \alpha \cdot |V|) \quad (12)$$

where $\alpha \in [0.01, 2.0]$ is the smoothing hyperparameter optimised via cross-validation.

3.3 Confidence-Threshold Gating and LLM Validation

The posterior confidence score for document T_i is defined as the gap between the top two class posteriors:

$$\kappa(X_i) = \max_{\alpha} P(C\alpha|X_i) - \text{second-max}_{\alpha} P(C\alpha|X_i) \quad (13)$$

Documents with $\kappa(X_i) < \tau$ are routed to FinBERT (Agent 3), which returns $F(T_i) = (p_+, p_-, p_0)$:

$$C_val(T_i) = \text{argmax}\{p_+, p_-, p_0\} \quad (14)$$

The resolved final class assignment is:

$$C^*(T_i) = \{ C_NB(T_i) \text{ if } \kappa(X_i) \geq \tau; C_val(T_i) \text{ if } \kappa(X_i) < \tau \} \quad (15)$$

3.4 Signal Fusion and Trading Rule

Agent 4 aggregates resolved assignments at each epoch t using source reliability weights $\omega_i \in (0,1]$:

$$S(t) = [\sum_i \omega_i \cdot \delta(C^*(T_i), C_+) - \sum_i \omega_i \cdot \delta(C^*(T_i), C_-)] / \sum_i \omega_i \quad (16)$$

The trading signal is generated by thresholding $S(t)$ against asymmetric boundaries θ_+ and θ_- :

$$\text{Signal}(t) = \{ +1 \text{ (Buy) if } S(t) \geq \theta_+; -1 \text{ (Sell) if } S(t) \leq \theta_-; 0 \text{ (Hold) otherwise } \} \quad (17)$$

3.5 Risk Management Filter

Realised volatility over the preceding 24-hour window (96 bars) is estimated as:

$$\sigma(t) = \sqrt{252 \cdot \text{std}\{\log(P_{\square}/P_{\square-1}) : t-H \leq s \leq t\}} \quad (18)$$

Volatility-scaled position size:

$$q(t) = \text{Signal}(t) \cdot \min(1, \sigma_{\text{target}} / \sigma(t)) \quad (19)$$

where $\sigma_{\text{target}} = 0.80$ matches the historical BTC volatility mean. A -20% drawdown circuit-breaker suspends trading with a 48-hour cooling period.

3.6 NB-AAI System Architecture

Agent 1: Data Collection Agent (DCA)

Role: Ingests raw text streams from Twitter/X, Reddit (r/Bitcoin, r/CryptoCurrency), and financial news APIs. Applies preprocessing: tokenisation, stop-word removal, stemming, emoji normalisation, deduplication.

Inputs: Raw text: Twitter API v2, Reddit PRAW, CryptoPanic/CoinDesk RSS

Outputs: Cleaned corpus $D(t)$, source reliability scores ω_i , vocabulary update $\Delta V(t)$

Agent 2: Naïve Bayes Classification Agent (NBCA)

Role: Applies Bernoulli Naïve Bayes classifier (Eqs. 10-11). Computes posterior probability and confidence score $\kappa(X_i)$. Routes high-confidence documents to Agent 4; low-confidence to Agent 3.

Inputs: Cleaned corpus $D(t)$, vocabulary V , trained parameters $\{P(C\alpha), P(w_a|C\alpha)\}$

Outputs: Class assignments $C_{\text{NB}}(T_i)$, confidence scores $\kappa(X_i)$, low-confidence subset $D_{\text{LC}}(t)$

Agent 3: LLM Validation Agent (LLMVA)

Role: Applies FinBERT to low-confidence subset $D_{\text{LC}}(t)$. Returns three-class softmax $F(T_i)=(p_+, p_-, p_0)$. Resolves final class $C^*(T_i)$ per Eq. (15). Proposes vocabulary updates for novel terms.

Inputs: Low-confidence document subset $D_{\text{LC}}(t)$ from Agent 2

Outputs: Validated class assignments $C_{\text{val}}(T_i)$, neutral abstentions, vocabulary update proposals ΔV

Agent 4: Signal Generation Agent (SGA)

Role: Aggregates resolved class assignments via source-weighted scoring (Eq. 16). Computes $S(t) \in [-1, +1]$. Applies threshold rule (Eq. 17) to produce $\text{Signal}(t) \in \{+1, 0, -1\}$.

Inputs: Resolved assignments $C^*(T_i)$, source weights ω_i , threshold parameters (θ_+, θ_-)

Outputs: Raw trading signal $\text{Signal}(t)$, aggregate sentiment score $S(t)$

Agent 5: Risk Management Agent (RMA)

Role: Applies volatility-scaled position sizing (Eqs. 18-19) and -20% maximum drawdown circuit-breaker. Enforces 50% single-asset exposure cap and -5% stop-loss per position.

Inputs: Raw signal $\text{Signal}(t)$, price history $P(t-H:t)$, current portfolio state

Outputs: Risk-adjusted position size $q(t) \in [-1, +1]$, order type, stop-loss levels

Agent 6: Orchestration Agent (OA)

Role: Coordinates full pipeline via LangGraph state machine. Monitors agent health, schedules execution, triggers vocabulary retraining when rolling accuracy drops below $\epsilon=0.60$, logs all decisions.

Inputs: System clock, agent status, rolling performance metrics

Outputs: Execution schedule, retraining triggers, performance logs, parameter updates

The system objective function maximises the validation Sharpe ratio over the hyperparameter vector $\theta = (\alpha, \tau, \theta_+, \theta_-, \sigma_{\text{target}}, \text{DD_max})$:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} SR_V(\theta) = \underset{\theta}{\operatorname{argmax}} [\mu_V(\theta) / \sigma_V(\theta)] \cdot \sqrt{252} \quad (20)$$

4. METHODOLOGY

4.1 Data Sources and Collection

Twitter/X	Twitter API v2 Academic Access. Query: (#Bitcoin OR #BTC OR #Ethereum OR #ETH) lang:en -is:retweet. Accounts $\geq$ 500 followers retained. Volume: ~2.1 million documents (Jan 2020–Dec 2024).
Reddit	PRAW API: r/Bitcoin, r/CryptoCurrency, r/ethtrader, r/CryptoMarkets. Posts/comments with $\geq$ 10 upvotes. Volume: ~890,000 documents. Provides longer-form discourse complementary to Twitter.
Crypto News	CryptoPanic API and CoinDesk RSS. Headlines and sub-headlines extracted. Volume: ~340,000 news items. Captures event-driven sentiment spikes.

Cryptocurrency OHLCV price data for BTC/USD and ETH/USD is sourced from Binance public REST API at 4-hour bar resolution (10,958 bars per asset). Transaction costs: 0.10% taker fee + 0.05% slippage per side (0.30% round-trip). Risk-free rate: 3-month US T-Bill yield from FRED (annualised average 3.2% over test period).

Table 2. Dataset Summary Statistics

Variable	BTC/USD	ETH/USD	Twitter	Reddit	News
Observations	10,958 bars	10,958 bars	2,100,000	890,000	340,000
Mean daily return	+0.31%	+0.38%	—	—	—
Ann. volatility	82.4%	96.7%	—	—	—
Max drawdown	−77.1%	−79.6%	—	—	—
Avg. docs/4hr bar	—	—	483	205	78

4.2 Preprocessing Pipeline

Six-step pipeline: (1) URL/mention removal; (2) emoji-to-text normalisation; (3) case folding and TweetTokenizer tokenisation; (4) stop-word removal (negation tokens retained); (5) Porter stemming; (6) near-duplicate deduplication (cosine similarity ≥ 0.95 collapsed).

Vocabulary sub-lexicons assigned via log-likelihood ratio:

$$LLR(w_a) = \log[P(w_a|C+) / P(w_a|C-)] \quad (21)$$

Labels generated from 4-hour forward returns:

$$y_i = \text{sign}(r_{\{t+4h\}} - \delta) \cdot \mathbb{1}\{|r_{\{t+4h\}}| > \delta\} \quad \text{where } \delta = 0.005 \quad (22)$$

4.3 Train-Validation-Test Split

Table 3. Temporal Data Split

Period	Window	Duration	Purpose	Regimes Covered
Training	Jan 2020–Dec 2020	12 months	Vocabulary, $P(w_a C\alpha)$ estimation	COVID crash, recovery
Validation	Jan 2021–Dec 2021	12 months	Hyperparameter θ^* optimisation	2021 bull run, correction
Test	Jan 2022–Dec 2024	36 months	Out-of-sample evaluation	All 4 regimes

Table 4. Market Regime Classification (Test Period)

Regime	Period	BTC Return	Characteristic
Regime 1: Crypto Winter	Jan–Dec 2022	−65.0%	Sustained bear; FTX collapse
Regime 2: Recovery I	Jan–Jun 2023	+82.5%	Gradual recovery; institutional re-entry
Regime 3: Recovery II	Jul–Dec 2023	+61.2%	ETF speculation; pre-halving momentum
Regime 4: Bull Run	Jan–Dec 2024	+123.8%	Bitcoin ETF approval; halving event

4.4 Agent Implementation Details

Agent 2 (NBCA): scikit-learn BernoulliNB, log-space inference, target latency < 2ms per document. Agent 3 (LLMVA): ProsusAI/finbert from HuggingFace, NVIDIA T4 GPU, FP16

mixed precision, batch size 32 (~8ms/doc batched). Agent 6 (OA): LangGraph state machine with six nodes connected by confidence-gate conditional edges. Walk-forward re-calibration at 6-month intervals during the test period.

Table 5. Baseline Models

Baseline	Description	Purpose
B1: Buy-and-Hold BTC	Passively hold BTC/USD	Market benchmark (H1)
B2: Buy-and-Hold ETH	Passively hold ETH/USD	Market benchmark (H1)
B3: Naïve Bayes Only	Agent 3 disabled; NB classification only	Quantifies LLM contribution (H2)
B4: FinBERT Only	All documents via FinBERT; no NB or agents 5-6	Quantifies NB contribution
B5: NB + Risk (No Orch.)	Agents 1,2,4,5 active; agents 3,6 disabled	Quantifies orchestration value (H3)

4.5 Evaluation Metrics

Directional accuracy, Sharpe ratio (annualised), maximum drawdown, Calmar ratio, and Sortino ratio. Sharpe annualisation:

$$SR = (\mu_p - r_f) / \sigma_p \cdot \sqrt{365 \times 6} \quad (23)$$

Statistical significance via Diebold-Mariano test (two-sided, 5% level):

$$DM = \mu_d / \sqrt{(\sigma_d^2 / T)} \sim N(0,1) \text{ under } H_0 \quad (24)$$

where $d = L(e^{\square_{\text{NB-AAI}}}) - L(e^{\square_{\text{baseline}}})$ is the loss differential using sign-error loss.

5. RESULTS AND DISCUSSION

5.1 Classification Performance

Table 6. Classification Performance on Test Set (Jan 2022 – Dec 2024)

Model	Dir. Acc (%)	Macro F1	F1 Bullish	F1 Bearish	F1 Neutral
B3: Naïve Bayes Only	58.3	0.541	0.563	0.512	0.548
B4: FinBERT Only	61.7	0.588	0.601	0.574	0.589
B5: NB + Risk (No Orch.)	60.1	0.567	0.581	0.548	0.572
NB-AAI (Full Pipeline)	66.4	0.641	0.658	0.619	0.646

Note: NB-AAI row highlighted. All NB-AAI vs. baseline differences significant at $p < 0.001$ (DM test).

NB-AAI achieves the highest directional accuracy of 66.4%, surpassing standalone Naïve Bayes by 8.1 percentage points and FinBERT-only by 4.7 points. The most pronounced improvement appears in bearish class F1 (0.619 vs. 0.512 for NB-only), confirming that LLM validation is especially valuable for sarcastic and negation-heavy content. The optimal confidence threshold $\tau^* = 0.28$, routing 21.3% of documents to FinBERT — within the 15-25% GPU utilisation target.

5.2 Trading Performance: Full Test Period

Table 7. Trading Performance — BTC/USD, Full Test Period (Jan 2022 – Dec 2024)

Model	Ann. Return (%)	Sharpe Ratio	Max DD (%)	Calmar Ratio	Sortino Ratio	# Trades
B1: Buy-and-Hold BTC	+18.4	0.31	−77.1	0.24	0.38	1
B3: Naïve Bayes Only	+29.7	0.74	−41.3	0.72	1.03	387
B4: FinBERT Only	+34.2	0.89	−38.6	0.89	1.21	312
B5: NB + Risk (No Orch.)	+31.8	0.83	−35.2	0.90	1.18	341
NB-AAI (Full Pipeline)	+52.6	1.47	−22.4	2.35	2.08	298

Note: NB-AAI highlighted in blue. DD = Drawdown. All Sharpe differences vs. NB-AAI significant at $p < 0.001$ (DM test).

NB-AAI achieves an annualised BTC Sharpe ratio of 1.47 and return of +52.6% — improvements of 98.6% and 65.2% over NB-only and FinBERT-only respectively. The Calmar ratio of 2.35 (vs. 0.24 for buy-and-hold) reflects nearly tenfold improvement in drawdown-adjusted returns. The lower trade count (298 vs. 387) reflects the risk management agent reducing spurious signal execution.

5.3 Regime-Stratified Performance

Table 8. NB-AAI vs. Buy-and-Hold Sharpe Ratio by Market Regime (BTC/USD)

Regime	Period	B1: Buy-Hold SR	NB-AAI SR	Improvement	DA (%)
Crypto Winter	Jan–Dec 2022	−0.48	0.82	+1.30	68.2
Recovery I	Jan–Jun 2023	0.71	1.38	+0.67	64.7
Recovery II	Jul–Dec 2023	0.89	1.52	+0.63	65.1
Bull Run 2024	Jan–Dec 2024	1.24	1.89	+0.65	63.9
Full Period	2022–2024	0.31	1.47	+1.16	66.4

Note: H3 confirmed — NB-AAI outperforms buy-and-hold across all four market regimes.

Outperformance is consistent across all four regimes. The largest gain occurs during the 2022 Crypto Winter ($\Delta SR = +1.30$), confirming that sentiment signals are most informative in bear markets where negative news propagates rapidly and predictably (Garcia & Schweitzer, 2015). The highest absolute Sharpe (1.89) is achieved in the 2024 Bull Run, driven by low prevailing volatility permitting larger positions under the σ_{target} constraint.

5.4 Ablation Study

Table 9. Ablation Study — Marginal Contribution of Each NB-AAI Component (BTC/USD)

Configuration	Sharpe Ratio	ΔSR vs. Full	Ann. Return (%)	Max DD (%)
Full NB-AAI Pipeline	1.47	—	+52.6	−22.4
Remove Agent 6 (Orchestration)	1.31	−0.16	+47.3	−26.1
Remove Agent 5 (Risk Mgmt)	0.98	−0.49	+51.8	−41.7
Remove Agent 3 (LLM Validation)	0.74	−0.73	+29.7	−41.3
Remove Agent 4 (Signal Fusion)	0.61	−0.86	+23.1	−48.6
Remove Agent 2 (Naïve Bayes)	0.89	−0.58	+34.2	−38.6
Remove Agents 3+5 (LLM+Risk)	0.51	−0.96	+22.4	−53.2
Remove Agents 3+5+6 (Core only)	0.38	−1.09	+17.8	−61.4

Note: Each row removes one component from the full pipeline. Removing agents 3+5+6 together degrades performance below all five baselines.

Key Ablation Finding — Superadditivity

LLM Validation (Agent 3) is the largest single contributor ($\Delta SR = -0.73$ when removed). Risk Management (Agent 5) is the primary drawdown protector (DD nearly doubles without it). Removing Agents 3+5 together reduces SR to 0.51 — below both NB-only (0.74) and FinBERT-only (0.89) baselines, confirming superadditivity: the NB-AAI architecture is more than the sum of its parts.

5.5 Statistical Significance Tests

Table 10. Diebold-Mariano Test Results (NB-AAI vs. Baselines, BTC/USD)

Comparison	DM Statistic	p-value	Significant?	Conclusion
NB-AAI vs. B3 (NB Only)	6.83	< 0.001	Yes ✓	H1 & H2 supported
NB-AAI vs. B4 (FinBERT Only)	3.41	0.001	Yes ✓	NB+Agents > LLM alone
NB-AAI vs. B5 (NB+Risk)	4.17	< 0.001	Yes ✓	H3 supported
NB-AAI vs. B3, ETH/USD	5.92	< 0.001	Yes ✓	H1 & H2 (ETH)
NB-AAI vs. B5, ETH/USD	3.76	< 0.001	Yes ✓	H3 (ETH)

Note: Two-sided DM test. H_0 : equal predictive accuracy. All tests reject H_0 at $p < 0.001$.

5.6 Discussion

Three principal theoretical implications emerge. First, the results demonstrate that the Naïve Bayes conditional independence assumption is not a fatal limitation when the classifier is embedded within a larger system compensating for its weaknesses. The confidence-gating mechanism converts Naïve Bayes into a computationally efficient first-pass filter, preserving its tractability while delegating failure modes to FinBERT — a hybrid architecture applicable beyond cryptocurrency trading. Second, the findings extend Kakushadze and Serur (2018) from a static single-model framework to a dynamic multi-agent system, maintaining theoretical continuity: the posterior confidence score $\kappa(X_i)$ from Eq. (13) uses the same Bayesian machinery to make principled routing decisions. Third, the regime-stratified results empirically support the asymmetric information transmission hypothesis from behavioural finance (De Long et al., 1990) — sentiment signals are most valuable in bear markets.

For practitioners, three actionable insights emerge: (1) the confidence-gating architecture reduces GPU inference costs by $\sim 79\%$ relative to a pure FinBERT approach while retaining 76% of the classification improvement; (2) source-reliability weighting provides a principled defence

against coordinated sentiment manipulation; (3) walk-forward re-calibration ensures practical deployability without full retraining.

5.7 Limitations

Four limitations warrant acknowledgement. First, results derive from backtested simulations rather than live trading; backtested performance is subject to optimism bias. Second, the study focuses exclusively on BTC and ETH; generalisability to smaller-capitalisation assets remains untested. Third, FinBERT was not fine-tuned on cryptocurrency-specific language. Fourth, regulatory risk, exchange downtime, and liquidity crises are not modelled.

Hypothesis Verdict:

H1 ✓ Supported

Naïve Bayes alone: DA = 58.3%, SR = 0.74 vs. buy-hold SR = 0.31 (DM $p < 0.001$). Naïve Bayes sentiment classification significantly outperforms passive buy-and-hold on both assets.

H2 ✓ Supported

LLM validation raises DA 58.3% $\rightarrow$ 66.4% and SR 0.74 $\rightarrow$ 1.47 on BTC (DM = 6.83, $p < 0.001$). False signal rate reduced 31.2%. LLM validation significantly improves Naïve Bayes.

H3 ✓ Supported

Full NB-AAI SR = 1.47 exceeds all baselines ($p < 0.001$). Improvement persists across all four market regimes including the 2022 Crypto Winter. Full pipeline is superadditive.

6. FUTURE RESEARCH DIRECTIONS

Six directions are identified, each motivated by a limitation or open question from the present study:

F1	Crypto-Specific Language Model Fine-Tuning Fine-tune FinBERT on a large corpus of Bitcoin/Ethereum social media, whitepapers, and exchange announcements to improve handling of domain-specific neologisms and community slang. Benchmark crypto-FinBERT against the general variant used here.
F2	Extension to Multi-Asset and DeFi Protocols Extend to Layer-1 protocols (Solana, Avalanche, Cardano) and DeFi tokens. Investigate whether an additional on-chain data agent monitoring wallet flows and DEX liquidity improves signal quality for lower-liquidity assets.
F3	Live Trading Deployment and Latency Optimisation Implement NB-AAI on a paper-trading infrastructure for ≥ 6 months before live capital deployment. Profile Agent 2 ($< 2\text{ms}$ target) and Agent 3 (batched GPU pipeline) under peak tweet volumes.
F4	Reinforcement Learning for Dynamic Threshold Optimisation Replace grid-search hyperparameter optimisation with a PPO or SAC RL agent that dynamically adjusts τ , θ_+ , θ_- , and α in response to recent market conditions and classification performance.
F5	Cross-Lingual Sentiment Expansion Integrate XLM-RoBERTa fine-tuned on multilingual financial text to capture Chinese, Korean, and Spanish cryptocurrency discourse, which may lead English-language signals by several hours.

F6	Interpretability and Regulatory Compliance Develop a SHAP-based post-hoc explanation layer over the vocabulary feature space to provide human-readable justifications for each trading signal — critical for regulatory compliance and institutional adoption.
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7. CONCLUSION

This paper introduced the Naïve Bayes Agentic AI (NB-AAI) framework — a novel hybrid architecture for sentiment-driven cryptocurrency trading that integrates the probabilistic interpretability of the Naïve Bayes Bernoulli classifier with the contextual sophistication of the FinBERT large language model and the adaptive real-time coordination of a six-agent orchestration pipeline. The framework extends the theoretical foundation of Kakushadze and Serur (2018), addressing three structural limitations of that static framework: vocabulary staleness, the independence assumption's failure on nuanced social media language, and the absence of adaptive multi-source signal coordination.

Evaluated on Bitcoin (BTC/USD) and Ethereum (ETH/USD) over the 36-month out-of-sample test period from January 2022 to December 2024 — encompassing the 2022 Crypto Winter, two recovery phases, and the 2024 bull run triggered by Bitcoin ETF approvals — the NB-AAI framework achieved an annualised BTC Sharpe ratio of 1.47, an annualised return of +52.6% net of transaction costs, and a maximum drawdown of −22.4%, compared to 0.31, +18.4%, and −77.1% for passive buy-and-hold. All improvements over five comparison baselines were confirmed statistically significant at the 1% level by the Diebold-Mariano test.

The ablation study revealed that NB-AAI exhibits superadditivity: the full six-agent pipeline outperforms the sum of individual component contributions. The LLM Validation Agent was identified as the largest classifier contributor ($\Delta\text{SR} = -0.73$ when removed); the Risk

Management Agent was the primary drawdown protector. All three research hypotheses (H1, H2, H3) were confirmed, with outperformance persisting across all four market regime conditions.

Beyond the empirical findings, this paper makes a methodological contribution by demonstrating that classical probabilistic models need not be discarded in the era of large language models. Instead, they serve as the interpretable, computationally efficient backbone of hybrid architectures in which LLMs are deployed selectively — as a precision instrument for marginal cases — rather than universally. This design philosophy may find application across a wide range of financial AI systems where interpretability, efficiency, and contextual richness must be balanced under real-time operational constraints.

The NB-AAI framework, its complete theoretical derivation, and its full experimental implementation are made available as open-source code to support replication and extension by the research community.

Summary of Original Contributions:

C1: NB-AAI Hybrid Framework

First integration of Naïve Bayes Bernoulli classification within a six-agent Agentic AI orchestration pipeline for cryptocurrency trading, grounded in Equations (5)-(20).

C2: Multi-Regime Empirical Evaluation

36-month out-of-sample evaluation across four market regimes (2022-2024). BTC Sharpe: 1.47 vs. 0.31 buy-hold; Calmar ratio: 2.35 vs. 0.24.

C3: Ablation Study with Marginal Attribution

Systematic six-component ablation quantifying each agent's marginal contribution. Confirms superadditivity of the full pipeline.

C4: Open-Source Reproducible Implementation

Python implementation with Docker container, fixed random seeds, and Colab notebooks. Full replication enabled.

REFERENCES

- [1] Abraham, J., Higdon, D., Nelson, J., & Ibarra, J. (2018). Cryptocurrency price prediction using tweet volumes and sentiment analysis. *SMU Data Science Review*, 1(3), 1-22.
- [2] Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.
- [3] Baur, D. G., Hong, K., & Lee, A. D. (2018). Bitcoin: Medium of exchange or speculative assets? *Journal of International Financial Markets, Institutions and Money*, 54, 177-189.
- [4] Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1), 1-8.
- [5] Corbet, S., Meegan, A., Larkin, C., Lucey, B., & Yarovaya, L. (2019). Exploring the dynamic relationships between cryptocurrencies and other financial assets. *Economics Letters*, 165, 28-34.
- [6] De Long, J. B., Shleifer, A., Summers, L. H., & Waldmann, R. J. (1990). Noise trader risk in financial markets. *Journal of Political Economy*, 98(4), 703-738.
- [7] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT 2019*, 4171-4186.
- [8] Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253-263.
- [9] Fang, F., Ventre, C., Basios, M., Kanthan, L., Martinez-Rego, D., Wu, F., & Li, L. (2022). Cryptocurrency trading: A comprehensive survey. *Financial Innovation*, 8(1), 1-59.
- [10] Garcia, D., & Schweitzer, F. (2015). Social signals and algorithmic trading of Bitcoin. *Royal Society Open Science*, 2(9), 150288.
- [11] Ji, Q., Bouri, E., Lau, C. K. M., & Roubaud, D. (2019). Dynamic connectedness and integration in cryptocurrency markets. *International Review of Financial Analysis*, 63, 257-272.
- [12] Jiang, Z., Xu, D., & Liang, J. (2017). A deep reinforcement learning framework for the financial portfolio management problem. *arXiv:1706.10059*.
- [13] Kakushadze, Z., & Serur, J. A. (2018). 151 Trading Strategies. Quantigic Solutions LLC / Springer Nature.

-
- [14] Khedr, A. M., Arif, I., El-Bannany, M., Alhashmi, S. M., & Sreedharan, M. (2021). Cryptocurrency price prediction using traditional statistical and machine learning techniques. *Intelligent Systems in Accounting, Finance and Management*, 28(1), 3-34.
- [15] Kim, J. W., & Han, I. (2000). Application of a hybrid genetic algorithm and neural network approach. *Expert Systems with Applications*, 19(3), 147-153.
- [16] Kraaijeveld, O., & De Smedt, J. (2020). The predictive power of public Twitter sentiment for forecasting cryptocurrency prices. *Journal of International Financial Markets, Institutions and Money*, 65, 101188.
- [17] Lamon, C., Nielsen, E., & Redondo, E. (2017). Cryptocurrency price prediction using news and social media sentiment. Stanford CS229 Technical Report.
- [18] LangChain. (2024). LangGraph: Stateful multi-agent workflows. <https://www.langchain.com/langgraph>.
- [19] Li, F., Hai, M., & Zhu, P. (2014). Sentiment analysis of financial news articles using performance indicators. *Knowledge-Based Systems*, 69, 14-23.
- [20] Li, Y., Wang, Z., & Chen, T. (2023). Multi-agent reinforcement learning for portfolio optimisation. *Expert Systems with Applications*, 210, 118599.
- [21] Lopez-Lira, A., & Tang, Y. (2023). Can ChatGPT forecast stock price movements? arXiv:2304.07619.
- [22] Loughran, T., & McDonald, B. (2011). When is a liability not a liability? *Journal of Finance*, 66(1), 35-65.
- [23] Manning, C. D., Raghavan, P., & Schutze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.
- [24] McNally, S., Roche, J., & Caton, S. (2018). Predicting the price of Bitcoin using machine learning. *IEEE EUROMICRO PDP 2018*, 339-343.
- [25] Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv:1301.3781.
- [26] Moura, J. (2024). CrewAI: Framework for orchestrating role-playing autonomous AI agents. GitHub. <https://github.com/crewAIInc/crewAI>.
- [27] Nadarajah, S., & Chu, J. (2017). On the inefficiency of Bitcoin. *Economics Letters*, 150, 6-9.
- [28] Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system. <https://bitcoin.org/bitcoin.pdf>.
- [29] Otabek, S., & Choi, J. (2024). Multi-level deep Q-networks for Bitcoin trading strategies. *Scientific Reports*, 14(1), 771.
- [30] Pang, B., Lee, L., & Vaithyanathan, S. (2002). Thumbs up? Sentiment classification using machine learning techniques. *EMNLP 2002*, 79-86.

-
- [31] Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behaviour. arXiv:2304.03442.
- [32] Porter, M. F. (1980). An algorithm for suffix stripping. *Program: Electronic Library and Information Systems*, 14(3), 130-137.
- [33] Rao, T., & Srivastava, S. (2012). Analyzing stock market movements using Twitter sentiment analysis. *ASONAM 2012*, 119-123.
- [34] Rish, I. (2001). An empirical study of the Naive Bayes classifier. *IJCAI Workshop on Empirical Methods in AI*, 41-46.
- [35] Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- [36] Schumaker, R. P., & Chen, H. (2009). Textual analysis of stock market prediction using breaking financial news. *ACM Transactions on Information Systems*, 27(2), 1-19.
- [37] Significant Gravitas. (2023). Auto-GPT. GitHub. <https://github.com/Significant-Gravitas/Auto-GPT>.
- [38] Smailovic, J., Grcar, M., Lavrac, N., & Znidarsic, M. (2014). Stream-based active learning for sentiment analysis in the financial domain. *Information Sciences*, 285, 181-203.
- [39] Sprenger, T. O., Tumasjan, A., Sandner, P. G., & Welpe, I. M. (2014). Tweets and trades: The information content of stock microblogs. *European Financial Management*, 20(5), 926-957.
- [40] Tetlock, P. C. (2007). Giving content to investor sentiment. *Journal of Finance*, 62(3), 1139-1168.
- [41] Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80-82.
- [42] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *NeurIPS 2017*, 5998-6008.
- [43] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS 2022*.
- [44] Wooldridge, M., & Jennings, N. R. (1995). Intelligent agents: Theory and practice. *Knowledge Engineering Review*, 10(2), 115-152.
- [45] Wu, Q., Bansal, G., Zhang, J., Wu, Y., Zhang, S., Zhu, E., Li, B., Jiang, L., Zhang, X., & Wang, C. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. arXiv:2308.08155.
- [46] Yang, Y., Uy, M. C. S., & Huang, A. (2020). FinBERT: A pretrained language model for financial communications. arXiv:2006.08097.
- [47] Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergising reasoning and acting in language models. *ICLR 2023*.

-
- [48] Zhang, H. (2004). The optimality of Naive Bayes. Proceedings of the 17th FLAIRS Conference. AAAI Press.
- [49] Zhang, W., Li, Y., & Chen, R. (2024). FinAgent: A multimodal foundation agent for financial trading. arXiv:2402.18485.
- [50] Zhang, X., Fuehres, H., & Gloor, P. A. (2011). Predicting stock market indicators through Twitter. *Procedia Social and Behavioral Sciences*, 26, 55-62.